

# Product Dissection for Zomato



## **Company Overview:**

Zomato is a leading food technology company headquartered in India, with a global presence in over 10,000 cities across 24 countries. Founded in 2008 by Deepinder Goyal and Pankaj Chaddah, Zomato started as a restaurant discovery and review platform, allowing users to explore dining options and read reviews before making a choice. Over the years, Zomato has evolved into a comprehensive food delivery and dining platform, offering services such as online food ordering, table reservations, and food delivery logistics.

## **Product Dissection and Real-World Problem Solved by zomato:**

**Convenience in Food Discovery:** Zomato addresses the need for easy exploration of restaurants and cuisines by providing a user-friendly platform with comprehensive restaurant listings, menus, and reviews.

**Efficient Delivery System:** It solves the challenge of time constraints and inconvenience in accessing food by offering a seamless food delivery service with quick turnaround times and real-time tracking.

**Diverse Food Options:** Zomato caters to various tastes and dietary preferences by offering a wide range of cuisines from local eateries to popular chains.

## 1. Time Constraints and Limited Access to Food:

- o **Real-World Challenge:** Busy lifestyles often lead to time constraints, making it challenging to cook meals or dine out.
- o **Zomato's Solution:** Zomato provides a convenient solution by partnering with numerous restaurants and offering quick delivery services, ensuring access to diverse culinary options even in areas with limited physical restaurants.

## 2. Food Discovery and Decision Fatigue:

- o **Real-World Challenge:** Choosing where to eat can be overwhelming due to the abundance of options.
- o **Zomato's Solution:** Zomato tackles this challenge through personalized recommendations, curated collections, and user reviews, helping users discover new dining options and make informed decisions quickly.

## 3. Lack of Transparency and Trust in Food Delivery:

- o **Real-World Challenge:** Concerns regarding hygiene & food safety, order accuracy, and delivery transparency can deter users from using food delivery services.
- o **Zomato's Solution:** Zomato ensures transparency and trust through rigorous partner vetting processes, real-time order tracking, and customer support mechanisms, fostering confidence among users.

## 4. Limited Scope of Restaurants to reach People:

- o **Real-World Challenge:** How can restaurants effectively reach a wider audience and increase their visibility?
- o **Zomato's Solution:** Zomato offers restaurants various marketing tools and features, including online listings, promotional campaigns, and targeted advertisements, helping them increase their visibility and reach a wider audience of potential customers.

## 5. Absence of Simple and Interactive Interface:

- o **Real-World Challenge:** How can users easily access and manage their dining preferences and history?
- o **Zomato's Innovative Solution:** Zomato's user-friendly interface and personalized features enable users to easily access and manage their dining preferences, order history, and saved restaurants, ensuring a seamless and personalized dining experience every time they use the platform.

In conclusion, Zomato's innovative approach to problem-solving in the real world has transformed the food industry landscape by addressing a myriad of challenges faced by consumers, restaurants, and food businesses. Through its comprehensive platform, global reach, and innovative technology solutions, Zomato has revolutionized the way people discover, order, and enjoy food. By fostering community engagement, empowering restaurants, prioritizing safety and hygiene, offering dynamic pricing and promotions, and continuously innovating, Zomato has positioned itself as a leader in the food technology space. With its commitment to enhancing the dining experience for all stakeholders, Zomato continues to shape the future of the food industry, making it more convenient, accessible, and enjoyable for everyone involved.

## Top Features of Zomato:

1. **Restaurant Discovery:** Zomato allows users to discover new restaurants based on location, cuisine, price range, and ratings, making it easy to find the perfect dining spot.
2. **User Reviews and Ratings:** Users can read reviews and ratings from other customers to make informed decisions about where to eat, ensuring a better dining experience.
3. **Online Food Ordering:** Zomato's online food ordering platform enables users to browse menus, place orders, and get food delivered from a wide range of restaurants, cafes, and eateries.
4. **Table Reservations:** Users can make table reservations at their favourite restaurants directly through the Zomato app or website, saving time and hassle.
5. **Real-time Order Tracking:** Zomato provides real-time tracking of food orders, allowing users to monitor the status of their delivery and know exactly when to expect their food.
6. **Hygiene Ratings:** Zomato assigns hygiene ratings to restaurants based on cleanliness and food safety standards, helping users make safer dining choices.
7. **Personalized Recommendations:** Zomato uses AI algorithms to provide personalized restaurant recommendations based on user preferences, order history, and dining behaviour.
8. **Multiple Payment Options:** Flexible payment methods including cash on delivery, digital wallets, and online banking.

9. **Deals and Discounts:** Zomato offers discounts, deals, and cashback incentives to users, making dining out or ordering in more affordable and rewarding.

10. **Social Features:** Users can engage with each other through social features such as liking, commenting, and sharing reviews, photos, and recommendations, creating a vibrant community of food enthusiasts.

**Impact on the Food Delivery Industry:** Zomato's innovative solutions to real-world problems in food delivery has not only provided immense convenience to consumers but also significantly influenced the food delivery industry by:

- **Expanding Restaurant Reach:** Partnering with Zomato has enabled restaurants to reach a broader customer base.
- **Creating Job Opportunities:** The demand for delivery personnel has increased, generating employment opportunities.
- **Cultural Exchange:** Zomato's platform promotes cultural diversity by facilitating the exploration and appreciation of various cuisines.

## Schema Description:

The schema for Zomato includes entities such as Users, Restaurants, Orders, Deliveries, Payments, and Items, each with specific attributes defining their properties and relationships with other entities.

### User Entity:

Users are fundamental to the platform, representing individuals who use the service:

- **UserID (Primary Key):** A unique identifier for each user.
- **Username:** The user's chosen username.
- **Gender:** Whether the user is Male or Female.
- **Email:** Email address used for account-related communications.
- **Phone\_Number:** The user's contact number.
- **Address:** Stored addresses for delivery purposes.
- **Password:** Encrypted password for account security.

### Restaurant Entity:

Restaurants are crucial partners, providing the diverse food options available on the platform:

- **RestaurantID (Primary Key):** A unique identifier for each restaurant.
- **Name:** Name of the restaurant.
- **Location:** Physical address of the restaurant.
- **Cuisine\_Type:** Types of cuisine offered.
- **Rating:** Average customer rating of the restaurant.

## Orders Entity:

Orders encapsulate the details of each transaction made by the users:

- **OrderID (Primary Key):** A unique identifier for each order.
- **UserID (Foreign Key referencing User Entity):** The user who placed the order.
- **RestaurantID (Foreign Key referencing Restaurant Entity):** The restaurant from which the order is placed.
- **Total\_Amount:** Total cost of the order.
- **Order\_Status:** Status of the order (e.g., preparing, on route).
- **Order\_Date:** The date and time when the order was placed.

## Delivery Entity:

Deliveries are essential for transporting orders from restaurants to users:

- **DeliveryID (Primary Key):** A unique identifier for each delivery.
- **OrderID (Foreign Key referencing Orders Entity):** The order being delivered.
- **Delivery\_ExecutiveID:** Identifier for the delivery executive.
- **Estimated\_Delivery\_Time:** Expected time for the order to be delivered.
- **Delivery\_Status:** Current status of the delivery (e.g., picked up, delivered).

## Payment Entity:

Payments record the financial transactions for each order:

- **PaymentID (Primary Key):** A unique identifier for each payment transaction.
- **OrderID (Foreign Key referencing Orders Entity):** The order for which the payment is made.
- **Amount:** Amount of the transaction.
- **Payment\_MethodID:** Method of payment (e.g., card, wallet, COD).
- **Payment\_Status:** Status of the payment (e.g., successful, pending).

## Payment Method Entity:

Payments record the financial transactions for each order:

- **PaymentID (Primary Key):** PaymentMethodID (Primary Key): Unique identifier for each payment method.
- **Method\_name:** Name of the payment method (e.g., card,upi, wallet, COD).

## Zomato\_Money Entity:

Zomato Money is an entity that represents the virtual wallet system integrated into the Zomato platform, allowing users to store funds for making payments seamlessly within the app. There are three sub categories under Zomato\_Money – 1. Zomato Wallet 2. Gift Card 3. Zomato Credit.

- **Category\_Name (Primary Key):** Name of all types of Zomato Money
- **UserID (Foreign Key referencing User Entity):** The user who have Zomato money.
- **Balance:** Amount that is present in each sub categories

## Item Entity:

Items represent the individual menu selections included in an order:

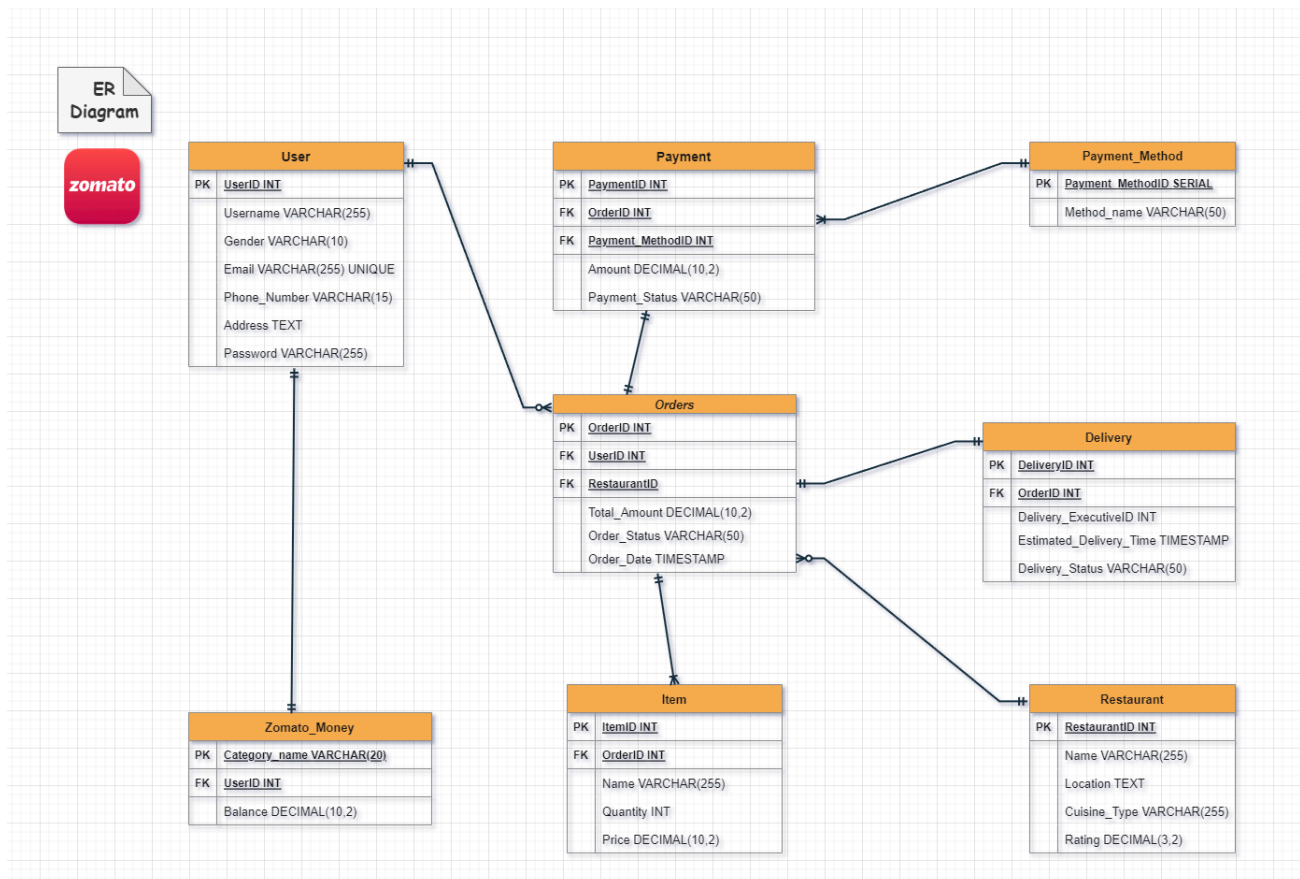
- **ItemID (Primary Key):** A unique identifier for each menu item.
- **OrderID (Foreign Key referencing Orders Entity):** The order including this item.
- **Name:** Name of the item.
- **Quantity:** Number of items ordered.
- **Price:** Price of the item.

## Relationships:

- **Users place Orders:** Each user can place zero, one and also multiple orders, and each order is placed by exactly one user. The cardinality is one mandatory-to-many optional (From Users to Orders).
- **Users have Zomato\_Money:** Each user have only one Zomato money in which they can hold balance. The cardinality is one-to-one (From users to Zomato Money)
- **Orders contain Items:** Each order can contain at least one or multiple items, and Each item is part of only one order [Once an item is ordered, it belongs to that specific order and is not associated with any other order.] The cardinality is one-to-many (From Order to Item).
- **Restaurants fulfil Orders:** Each order is associated with one restaurant, and a restaurant can have zero or multiple orders. The cardinality is many-to-one (From Order to Restaurant).
- **Orders require Delivery:** Each order is linked to a single delivery instance, and each delivery is associated with one order. The cardinality is one-to-one (From Order to Delivery).
- **Orders involve Payment:** One Order is associated with one Payment. One Payment corresponds to one Order. The cardinality is one-to-one (From Order to Payment).
- **Payment Method used for making a payment:** Each payment is associated with one payment method. And One payment method can be used for multiple payments. The cardinality is one-to-one (From users to Zomato Money)

# Entity Relationship Diagram

Let's create an Entity-Relationship (ER) diagram to clearly illustrate the connections and characteristics of the elements in the Swiggy database structure. This ER diagram will act as a visual guide, highlighting the essential parts of Swiggy's data architecture. Utilizing this diagram will help us understand the complex relationships and interactions that shape the functioning of the platform.



## Conclusion

In this study, we have analysed Zomato's operational framework and its significant impact on the food delivery ecosystem. Zomato has revolutionized the way food is ordered and delivered by leveraging technological advancements and prioritizing user-centric solutions. The platform incorporates various entities such as users, restaurants, orders, deliveries, payments, and menu items, establishing a robust and efficient system. This comprehensive structure not only facilitates seamless transactions but also accommodates diverse consumer needs and preferences effectively.

By examining Zomato's model, we gain valuable insights into how it effectively addresses the challenges within the food delivery sector, providing convenience to customers while offering crucial support to restaurant partners. Zomato's strategic initiatives and adaptable technology have propelled its growth, solidifying its position as a key player in the dynamic landscape of online food services.

Individual Contribution – Suman Sadhukhan

Video Presentation – <https://youtu.be/qyDPK8mrtYQ>

GitHub Link – [https://github.com/techysuman27/Product\\_Dissection\\_Zomato\\_Relational\\_Database](https://github.com/techysuman27/Product_Dissection_Zomato_Relational_Database)